Cultural Matching and the Persistence of Social Ties

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Introduction

A central puzzle in the sociology of culture and social networks is the co-evolution of social ties and cultural tastes. Traditional sociological models of taste formation posit that an individual's cultural profile is largely a product of their current social network configuration. Tastes are assumed to diffuse through existing homophilous ties, which are themselves structured by socio-demographic similarity—"status homophily" ((Lazarsfeld and Merton 1954; McPherson et al. 2001)). Under this view, "birds of a feather flock together," and due to this mutual flocking, they end up consuming the same cultural goods ((Mark 1998b, 2003)). Tastes are construed as highly malleable, "thin" contents liable to shift whenever the surrounding network ties change.

Yet, longitudinal network research reveals that personal networks are constantly in flux, with ties constantly being added, modified, and deleted ((Burt 2000, 2002; Wellman et al. 1997)). This empirical fact of network volatility raises a major question: if networks are unstable and tastes are merely thin, network-contingent contents, how do we account for the durable, resilient cultural profiles that individuals maintain over time ((Bourdieu 1984; Holt 1998))?

In this paper, we develop and test a **cultural matching model** of tie persistence. We propose that instead of tastes merely adapting to match pre-existing networks, pre-existing cultural dispositions serve as an active "social glue" that determines which network ties survive and which dissolve. Under this model, dyadic cultural matching (commonalities in tastes and activities) increases the over-time stability and cognitive salience of social ties, net of usual dyadic determinants such as subjective closeness and tie duration.

Using rich longitudinal ego-network data from the NetSense study, we track the persistence of social ties across multiple college-year transitions. To isolate peer dynamics, we restrict our focus to non-kin, on-campus student ties. We unpack this cultural matching mechanism in two ways. First, we establish the baseline protective effect of aggregate cultural matching on tie persistence. Second, we explore the role of "network opacity"—the degree to which an ego lacks knowledge of their alter's preferences (indicated by "Don't Know" responses)—as an independent predictor of tie decay.

Theoretical Framework

Tastes and Social Networks: Two Perspectives

The relationship between cultural tastes and social networks has been conceptualized in two contrasting ways in sociological theory. The **network transmission perspective** views tastes as malleable, network-contingent contents ((Erickson 1988, 1996; Mark 1998a, 2003)). In this view, social ties are the "pipes" or conduits through which cultural practices diffuse ((Borgatti 2005; Carley 1991)). Tastes are learned, adopted, and abandoned based on local bandwagon imitation within an actor's immediate ego network ((Mark 2003)). Because personal networks exhibit substantial over-time volatility—with 30% to 60% turnover annually in most personal networks ((Morgan et al. 1997; Suitor et al. 1997))—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the **cultural capital perspective** conceptualizes tastes as highly durable, embodied dispositions reaped from early socialization ((Bourdieu 1984, 1990; Holt 1997, 1998)). Tastes are part of a person's embodied cultural capital, encoded in memory traces, habitual tendencies, and cognitive schemas that are highly resilient and resistant to short-term network fluctuations ((Bourdieu 2000; Lizardo 2006)). If tastes are durable and personal networks are volatile, it is highly probable that tastes serve as *drivers* of network connections rather than being exclusively determined by them ((Lizardo 2006)).

This leads to the concept of **choice homophily** or cultural matching: new social connections are selected, and existing connections are maintained, based on pre-existing cultural compatibility ((Lazarsfeld and Merton 1954; Werner and Parmelee 1979)). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact selection keyed around cultural alignment, rather than homophily acting as an exogenous structural constraint ((Carley 1991; DiMaggio 1987)).

Cultural Matching and Tie Persistence

A major challenge for any newly formed social relationship is what Burt ((Burt 2000)), drawing on Stinchcombe ((Stinchcombe 1965)), refers to as the **"liability of newness."** Newly formed dyadic ties are fragile and exhibit a high baseline hazard of decay ("tie decay"), which steeply declines as a relationship matures and ripens into close friendship. What stabilizes a tie and helps it overcome this liability of newness?

While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we propose that **cultural matching** serves as a vital stabilizing mechanism. Shared tastes and cultural practices provide a common cognitive and behavioral baseline that facilitates smooth, low-cost social interaction ((Collins 2004; DiMaggio 1987)). When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the ego's active network ((Burt 2000)).

Therefore, we expect that commonalities in cultural tastes and leisure practices will significantly reduce the hazard of tie decay, net of subjective closeness and structural controls (**The Baseline Matching Hypothesis**).

Network Opacity

Finally, we introduce the concept of **network opacity**. If shared cultural knowledge stabilizes ties, then a lack of mutual cultural knowledge—network opacity—is a diagnostic signal of relationship precarity. When an ego reports "Don't Know" regarding an alter's preferences across multiple cultural domains, this indicates a lack of conversational intimacy and shared experiences, which should independently predict a higher hazard of tie dissolution ((Burt 2002; Wellman et al. 1997)). We test both the main effect of network opacity on tie decay and whether opacity moderates the protective effect of known cultural matches.

Data and Methods

Data for this analysis comes from the NetSense study, a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students' social networks, demographic backgrounds, and cultural preferences. A comprehensive codebook, data dictionary, and repository for the NetSense project can be found at https://olizardo.github.io/NetSense-Codebook/.

We collected information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended activity nominations. For the closed-form items, egos were asked to rate both their own and their alters' level of interest across six broad cultural and leisure domains: music, movies, books, sports, games, and outdoor activities. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity.

Study Waves

The NetSense study is a longitudinal survey of college students collected over multiple waves. For this analysis of tie persistence, we pool data across adjacent waves to construct a person-period dataset. To isolate peer dynamics, we restrict our sample to non-kin, on-campus ties (i.e., alters who are also students at Notre Dame). Specifically, we focus on transitions between: - **Wave 1 to Wave 2**: Capturing tie dynamics over the first year of college. - **Wave 2 to**

Wave 5: Spanning the middle years of the college experience (approximately a 3-year gap). - **Wave 5 to Wave 6**: Capturing the transition through the final year and toward graduation.

Measures

Dependent Variable - *Tie Persistence*: A binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2), and 0 if the tie has dissolved (i.e., the alter is no longer listed in the ego's network).

Independent Variables - *Closed-Form Cultural Matching*: A count variable (ranging from 0 to 6) indicating the number of closed-form cultural domains in which the ego and alter share similar tastes or participation levels. To measure ego preferences, respondents were asked, "Please indicate how much you enjoy [listening to music / watching movies / reading books / following sports / playing games / outdoor activities]," with response options: "Very much," "Somewhat," or "Not that much." To measure alter preferences, egos were asked, "Please indicate how much you think each of your contacts enjoys [listening to music / watching movies / reading books / following sports / playing games / outdoor activities]," with response options: "Very much," "Somewhat," "Not that much," or "Don't know." A match is defined as both the ego and the alter having positive engagement (e.g., "Very much" or "Somewhat") in a given domain. - *Open-Ended Activity Matching*: A count variable (ranging from 0 to 5) capturing the number of shared open-ended activity nominations. Egos were first asked, "Please list a total of five activities that you enjoy doing. Examples of activities could be bicycle riding, playing poker, watching soap operas, weight lifting..." For each nominated activity, egos were subsequently asked to indicate whether their alter also engaged in it. - *Cultural Network Opacity*: A count variable (ranging from 0 to 6) measuring the number of cultural domains for which the ego responded "Don't know" regarding their alter's preferences on the closed-form cultural items described above.

Control Variables - *Tie Characteristics*: We control for the subjective closeness of the relationship based on the question, "Please indicate the option that best describes your relationship with each person." The responses were coded into three categories: `Not Close` ("...less than close in the sense that you don't mind interacting with the person, but you have no wish to develop a friendship" or "...distant in the sense that you really don't enjoy spending time with the person unless it is necessary"), `Somewhat Close` ("...merely close, in the sense that you enjoy the person, but don't count him or her among your closest personal contacts"), and `Close` ("...especially close in the sense that this is one of your closest personal contacts"). We control for the duration of the tie, based on the question, "Approximately, how long in years have you known each person?" We also control for formal relationship type using the question, "What is their relationship to you?" From the response categories (Parent, Sibling, Other family, Significant other, Co-worker, Friend, Acquaintance, Other, Roommate/Dormmate), we created dummy variables indicating whether the tie is a `Friend` or a `Roommate/Dormmate`. - *Frequency of Interaction*: We control for how often the ego and alter interact, asking "On average, how frequently did you interact with each person? Please include phone call, text and email interactions as well as face-to-face interactions." Responses are categorized as `Daily`, `Weekly`, `Monthly`, with `Less often` serving as the reference category. - *Demographics*: Ego and alter gender (`Women / Men`), based on the question, "Please indicate the gender of each of your contacts." We include an interaction between ego and alter gender to account for gender homophily. We also control for race homophily (`Homophilous`, `Heterophilous`, `Unknown`) derived from demographic surveys of the ego and alter. - *Time Period*: Dummy variables for the specific wave intervals (Wave 1-2, Wave 2-5, Wave 5-6) to account for differing baseline rates of tie dissolution over different time gaps.

For a full summary of descriptive statistics for all variables used in the analysis, including means, standard deviations, and percentages, see Tables 4 and 5 in the Appendix.

Methodological Approach and Modeling Strategy

We adopt a unified **Discrete-Time Survival Analysis** (event history modeling) approach. To accomplish this, we structured the longitudinal network data into a person-period format, pooling all adjacent wave transitions (Wave 1 to 2, Wave 2 to 5, and Wave 5 to 6) into a single dataset. We model the probability of tie persistence (vs. dissolution) as a function of cultural matching and network opacity using mixed-effects logistic regression (`lme4::glmer` in R).

This updated modeling strategy introduces several key improvements: 1. **Simultaneous Estimation**: By modeling all waves simultaneously, we gain statistical power and provide a generalized estimate of tie persistence over the entire study period. 2. **Baseline Hazard Modeling**: We include fixed effects for the specific wave periods (`period`) to flexibly control for the baseline hazard of tie dissolution, acknowledging that tie decay rates differ between the 1-year (Wave 1-2, Wave 5-6) and 3-year (Wave 2-5) intervals. 3. **Accounting for Clustering**: We include a random intercept for each ego (`1 | egoid`) to account for the unobserved heterogeneity and non-independence of multiple alter ties nested within the same ego. 4. **Predictive Margins**: To facilitate interpretation of the logistic regression coefficients, we use the `marginaleffects` package to compute and visualize predicted probabilities of tie persistence across the range of our variables of interest.

Results

Main Effects Models

The following models assess the main effects of closed-form cultural matching, open-ended cultural matching, and network opacity on the odds of tie persistence. Model 1 shows that closed-form matches independently predict persistence. Model 2 introduces open-ended matches, demonstrating the value of incorporating both open and closed-form items. Model 3 incorporates network opacity (the number of "Don't know" responses regarding an alter's preferences). The main effects model reveals that a lack of shared cultural knowledge signals a weaker tie.

Odds Ratios for Tie Persistence (Main Effects Models)

	Model 1: Closed	Model 2: + Open	Model 3: + Opacity
Closed-Form Matches	1.225***	1.174**	
	(0.060)	(0.059)	
Open-Ended Matches		1.322***	1.310***
		(0.072)	(0.071)
Network Opacity			0.748***
			(0.042)
Subjective Closeness: Somewhat	0.526***	0.574***	0.583***
	(0.073)	(0.081)	(0.082)
Subjective Closeness: Close	0.332***	0.365***	0.419***
	(0.075)	(0.083)	(0.097)
Tie Duration (Years)	1.026	1.015	0.987
	(0.048)	(0.048)	(0.048)
Tie Duration Squared	0.998	0.999	1.000
	(0.003)	(0.003)	(0.003)
Ego: Woman	1.084	1.045	1.016
	(0.404)	(0.392)	(0.385)
Alter: Woman	1.414*	1.300	1.247
	(0.240)	(0.224)	(0.217)
Ego Woman × Alter Woman	0.622+	0.714	0.750
	(0.163)	(0.190)	(0.202)
Roommate/Dormmate	1.117	1.031	0.886
	(0.441)	(0.413)	(0.357)
Friend	0.811	0.775	0.727
	(0.224)	(0.217)	(0.204)
Frequency: Weekly/Daily	1.120	1.158	1.115
	(0.184)	(0.192)	(0.187)
Race Homophily: Same Race	0.967	0.961	0.921
	(0.191)	(0.191)	(0.185)
Race Homophily: Unknown	0.162***	0.165***	0.157***
	(0.030)	(0.031)	(0.030)

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Odds Ratios for Tie Persistence (Main Effects Models) (Continued)

Time Period: Wave 2-5	0.053***	0.074***	0.063***
	(0.018)	(0.025)	(0.022)
Time Period: Wave 5-6	0.090***	0.120***	0.088***
	(0.031)	(0.042)	(0.032)
Num.Obs.	3251	3251	3251
AIC	2562.3	2536.7	2518.5
BIC	2665.8	2646.3	2628.1

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Marginal Effects

Figure Figure 1 visualizes the marginal effect (calculated as the change from the minimum to the maximum value) of closed-form matches, open-ended matches, and network opacity on the predicted probability of tie persistence.

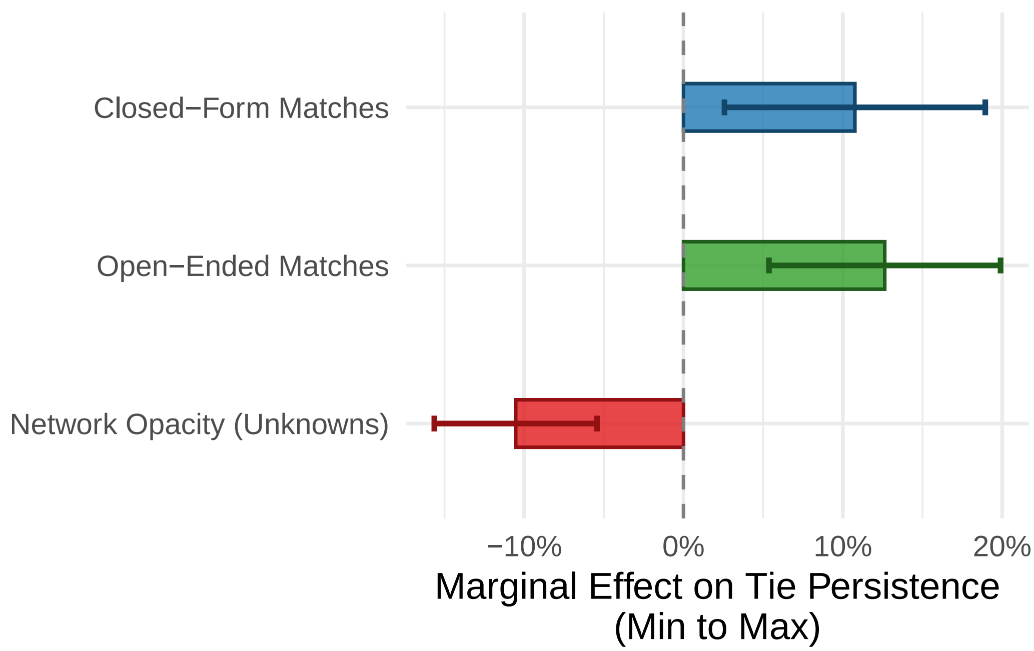

Figure 1: Marginal Effects of Cultural Matches and Network Opacity (Min to Max)

Subjective Closeness Interactions

To test whether the protective effects of cultural matching or the detrimental effects of network opacity depend on the strength of the tie, we estimated auxiliary interaction models between subjective closeness and each of the three cultural variables. While interaction terms in logistic regression models can be difficult to interpret directly from coefficients due to the non-linear link function, computing predicted probabilities across the range of the variables provides a clearer picture of the underlying dynamics.

Figure Figure 2 plots the marginal effects of the three cultural variables across the three levels of subjective closeness.

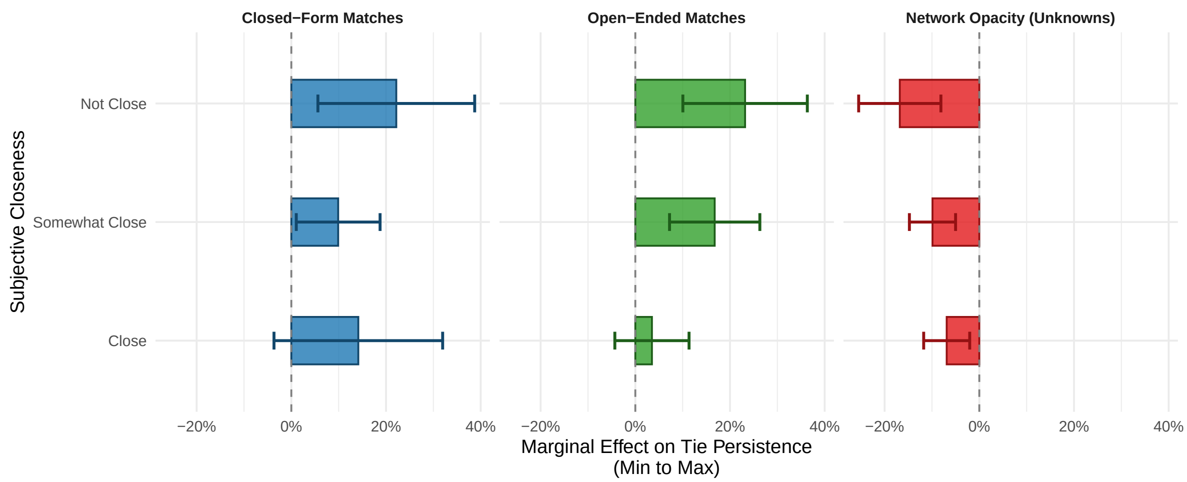

Figure 2: Predicted Probability of Tie Persistence by Cultural Matching, Network Opacity, and Subjective Closeness

As the marginal effects plot clearly demonstrates, the role of cultural matching varies substantially by subjective closeness. For “Not Close” ties (the steepest slopes), an increase in shared cultural tastes—whether measured through closed-form domains or open-ended activities—dramatically increases the probability of tie persistence. Conversely, for ties that are already “Close,” the slope is essentially flat. This suggests that shared cultural tastes provide critical scaffolding that helps weak ties survive, whereas strong ties are resilient to decay regardless of cultural matching. Similarly, network opacity exerts its most severe negative effect on ties that are not subjectively close. ## Appendix: Descriptive Statistics

Table 1: Descriptive Statistics for Continuous Variables

Variable	Mean	SD	Min	Max
Closed-Form Matches	2.12	1.47	0.00	6
Open-Ended Matches	2.44	1.32	0.00	5

Table 1: Descriptive Statistics for Continuous Variables

Variable	Mean	SD	Min	Max
Cultural Network Opacity	0.81	1.51	0.00	6
Tie Duration (Years)	3.01	4.19	0.08	20

Table 2: Descriptive Statistics for Categorical Variables

Variable	Category	N	Percent
Alter Gender	Men	1160	53.5
Alter Gender	Women	1007	46.5
Ego Gender	Men	1127	52.7
Ego Gender	Women	1010	47.3
Friend	No	236	10.9
Friend	Yes	1939	89.1
Race Homophily	Heterophilous	217	10.0
Race Homophily	Homophilous	500	23.0
Race Homophily	Unknown	1458	67.0
Roommate/Dormmate	No	2062	94.8
Roommate/Dormmate	Yes	113	5.2
Subjective Closeness	Close	302	14.1
Subjective Closeness	Not Close	712	33.3
Subjective Closeness	Somewhat Close	1127	52.6
Tie Persisted	No	1489	68.5
Tie Persisted	Yes	686	31.5
Time Period	Wave 1 to 2	104	4.8
Time Period	Wave 2 to 5	1365	62.8
Time Period	Wave 5 to 6	706	32.5